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SpeedFlower: a comprehensive speed breeding protocol for indica and japonica rice

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To increase rice yields and feed billions of people, it is essential to enhance genetic gains. However, the development of new varieties is often hindered by longer generation times and seasonal constraints. To address these limitations, a speed breeding facility has been established and new and robust protocol has been developed. The latest generation of speed breeding protocol allows to grow 4 to 5 generations of indica and/or japonica rice in a year. Our findings reveal that a high red-to-blue (2R>1B) ratio of the spectrum, followed by green, yellow, and far-red light; a 24-hour long day (LD) photoperiod for the initial 15 days of the vegetative phase followed by 10 h in the later stage; and 32/30°C day and night temperatures and 65% humidity facilitate early flowering ranging from 52 to 60 days at high light intensity (800 µmol m $^{-2}$ s^{-1}). Furthermore, the use of prematurely harvested seeds and gibberellic acid treatment reduced the maturity duration by 50%. Further, the developed protocol was validated on a diverse set of 198 rice accessions from 3K RGP subset encompassing all 12 distinct groups of Oryza sativa L. classes. Our results confirmed that one generation can be achieved using the developed protocol within 58 to 71 days, resulting in 5.1 to 6.3 generations per year across the 12 sub-groups. This breakthrough enables us to enhance genetic gain, which could feed half of the world's population dependent on rice. This demonstrates the robustness and effectiveness of the protocol in achieving accelerated rice breeding cycles.